

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 March 2023

22MCA103 - COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit - I		Marks	BT*	CO*	PO*
1. a)	Convert the following numbers: i) $BCA01.BA(16) = (?)_8$ ii) $1011.1101(10) = (?)_2$	4	L2	1	2
b)	Express the following as a function of max-terms: i) $F(X,Y,Z) = (X+Y)(X+Y')Z'$ ii) $F(X, Y, Z) = (XY + Z)(Y + XZ)$	5	L3	1	2
c)	Find the duality and complement of following expression: $((ABD)' + (A'BC) \cdot (AC'D'))' + (AB' + D')) ACD$	5	L3	1	2
d)	Implement XNOR and XOR gates using only NOR gates.	6	L3	1	2
2. a)	Express the following function in a SOP and POS form: i) $A \odot B \oplus C \oplus D$ ii) $F(A, B, C, D) = \sum(0, 2, 3, 5, 7)$	8	L3	1	2
b)	Prove the following Boolean functions: i) $(A + C)(AD + AD') + AC + C = A + C$ ii) $A'(A + B) + (B + AA)(A + B') = A + B$ iii) $A'B'C + A'BC' + A'BC + AB'C' + AB'C + ABC' + ABC = A + B + C$	12	L3	1	2
Unit - II					
3. a)	Implement $F = m(1, 3, 5, 6)$ using 4×1 MUX.	5	L3	2	2
b)	Define De-Multiplexer. Implement 1:4 DEMUX with basic gates.	7	L3	2	2
c)	Simplify the following Boolean functions by using the K-Map and show the simplified SOP and POS form $Y = F(A, B, C, D) = A'BCD + D' + C'D' + BCD + d(1, 3)$	8	L2	2	2
4. a)	Design the circuit for octal to binary encoder.	8	L3	2	2
b)	Simplify Boolean expression: $F(A, B, C, D) = \sum(4, 5, 6, 8, 9, 10, 13) + d(0, 7, 15)$ using tabulation method and find essential prime implicants.	12	L3	2	2
Unit - III					
5. a)	Illustrate 2's Complement Adder/Subtraction operation with the help of a logic diagram.	10	L3	3	2
b)	Draw the circuit of sequential binary multiplier and multiply two numbers $M=1101$, $Q=1011$ using sequential binary multiplier.	10	L2	3	2
6. a)	Multiply two numbers $A = +21$, $B = -18$ using Booth's algorithm.	4	L2	3	1
b)	Implement Full Subtractor using NAND Gates.	8	L3	3	2
c)	Perform the following operations using 2's complement Arithmetic: i) $(28) - (96)$, ii) $(30) - (5)$, iii) $(20) + (-27)$, iv) $(-17) + (9)$	8	L3	3	2
Unit - IV					
7. a)	Register R1 and R2 of a computer contain the decimal values 1200 and 4600. What is the effective address of the memory operand in each of the following instructions? i) Load 20(R1), R2 ii) Move #1000, R2 iii) Add -(R1), R2 iv) Add R1, R2	4	L2	4	1
b)	Show how the operation $Z = A + B - C \cdot D$ can be implemented in a single and double accumulator processor.	6	L3	4	2

P.T.O.

- c) Explain the operations performed by the following instructions:
- | | | | | |
|-------------------|--------------------|----|----|-----|
| i) LOAD LOCA, R1 | ii) ADD 1000, R5 | | | |
| iii) MOVE #10, R0 | iv) STORE R1, LOCB | | | |
| v) ADD (R2), R0 | | 10 | L3 | 4 2 |
- 8 a) Write an assembly language program to add 10 numbers stored in consecutive memory locations.
- | | | | | |
|--|---|----|---|---|
| | 8 | L2 | 4 | 2 |
|--|---|----|---|---|
- b) Explain the various addressing modes in detail.
- | | | | | |
|--|----|----|---|---|
| | 12 | L2 | 4 | 2 |
|--|----|----|---|---|

Unit - V

- 9 a) Explain the operation of ROM with a neat diagram. Also explain the different types of ROM.
- | | | | | |
|--|----|----|---|---|
| | 10 | L2 | 5 | 1 |
|--|----|----|---|---|
- b) Explain the two methods of bus arbitration.
- | | | | | |
|--|----|----|---|---|
| | 10 | L2 | 5 | 1 |
|--|----|----|---|---|
10. a) Define Interrupt. Briefly explain the operation of a hardware interrupt with a neat diagram.
- | | | | | |
|--|----|----|---|---|
| | 10 | L1 | 5 | 1 |
|--|----|----|---|---|
- b) A byte addressable computer has a small data cache capable of holding Eight 32 bit words. Each cache block consists of One 32 bit word. When a given program is executed the processor reads data from the following sequence of hex addresses.
200, 204, 208, 20D, 214, 210, 200, 204, 218, 21D, 24D, 214
this pattern is repeated for 4 times.
Show the contents of the cache at the end of each pass through this loop for an associative mapped cache that use the LRU replacement algorithm. Compute hit ratio. Assume that the cache is initially empty.
- | | | | | |
|--|----|----|---|---|
| | 10 | L1 | 5 | 1 |
|--|----|----|---|---|

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
